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PRACTICAL. No: 1	**Socket Programming – Client-Server Communication using Python**

AIM:

To implement a two-way chat communication between a server and a client using Python's socket programming.

CODE:

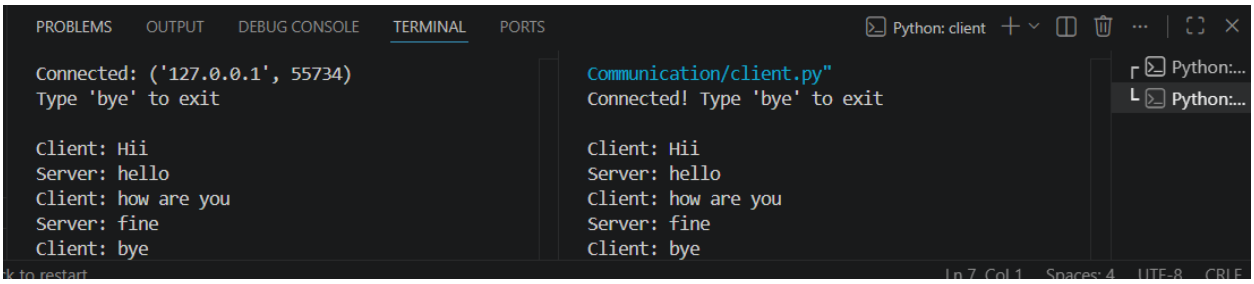
SERVER.PY

```
import socket
server = socket.socket()
server.bind(('localhost', 5000))
server.listen(1)
print("Waiting for connection...")
conn, addr = server.accept()
print(f"Connected: {addr}\nType 'bye' to exit\n")
while True:
    msg = conn.recv(1024).decode()
    print("Client:", msg)
    if msg.lower() == 'bye': break
    reply = input("Server: ")
    conn.send(reply.encode())
    if reply.lower() == 'bye': break
conn.close()
server.close()
```

CLIENT.PY

```
import socket
client = socket.socket()
client.connect(('localhost', 5000))
print("Connected! Type 'bye' to exit\n")
while True:
    msg = input("Client: ")
    client.send(msg.encode())
    if msg.lower() == 'bye': break
    reply = client.recv(1024).decode()
    print("Server:", reply)
    if reply.lower() == 'bye': break
client.close()
```

OUTPUT:



RESULT

Thus, the simple client-server communication program using socket programming in Python was implemented successfully and messages were exchanged between server and client.

PRACTICAL. No: 2	RMI / Simulate RPC in Python

AIM:-

To simulate Remote Procedure Call (RPC) in Python using socket programming and perform arithmetic operations between client and server.

CODE:

```
import socket
server = socket.socket()
server.bind(('localhost', 5000))
server.listen(1)
print("RPC Server Waiting...")
while True:
    conn, addr = server.accept()
    print("Connected:", addr)
    data = conn.recv(1024).decode()
    if data.lower() == "bye":
        print("Server Closed")
        conn.close()
        break
    parts = data.split()
    op = parts[0]
    a = int(parts[1])
    b = int(parts[2])
    if op == "add":
        result = a + b
    elif op == "sub":
        result = a - b
```

```
    elif op == "mul":
        result = a * b
    elif op == "div":
        result = a / b
    else:
        result = "Invalid Operation"
    conn.send(str(result).encode())
    conn.close()
server.close()
```

CLIENT.PY

```
import socket
while True:
    client = socket.socket()
    client.connect(('localhost', 5000))
    op = input("Enter operation (add/sub/mul/div) or bye: ")
    if op.lower() == "bye":
        client.send("bye".encode())
        client.close()
        break
    a = input("Enter first number: ")
    b = input("Enter second number: ")
    msg = op + " " + a + " " + b
    client.send(msg.encode())
    result = client.recv(1024).decode()
    print("Result:", result)
    client.close()
```

OUTPUT:

```
/Users/R.BAVADHARANI/OneDrive/Desktop/Distributed System Lab Experiments/RPC-PY
THON/server.py"
RPC Server Waiting...
Connected: ('127.0.0.1', 58781)
Connected: ('127.0.0.1', 58784)
```

```
THON/client.py"
Enter operation (add/sub/mul/div) or bye: add
Enter first number: 50
Enter second number: 45
Result: 95
Enter operation (add/sub/mul/div) or bye:
```

RESULT:-

Thus, the program to simulate Remote Procedure Call (RPC) in Python was implemented successfully and the arithmetic operation was executed remotely.

PRACTICAL. No: 3	Message Passing Between Two Processes using Python

AIM:

To demonstrate message passing between two processes using Python multiprocessing and queue.

CODE:

```
from multiprocessing import Process, Queue
def sender(q, msg):
    q.put(msg)
    print("Message Sent")
def receiver(q):
    msg = q.get()
    print("Message Received:", msg)
if __name__ == "__main__":
    q = Queue()
    msg = input("Enter message: ")
    p1 = Process(target=sender, args=(q, msg))
    p2 = Process(target=receiver, args=(q,))
    p1.start()
    p1.join()
    p2.start()
    p2.join()
```

OUTPUT:

```
PS C:\Users\R.BAVADHARANI\OneDrive\Desktop\Distributed System Lab Experiments> & C:/Users/R.BAVADHARANI/AppDat
/OneDrive/Desktop/Distributed System Lab Experiments/Message-Passes-two-processes/process.py"
Enter message: HELLOO
Message Sent
Message Received: HELLOO
PS C:\Users\R.BAVADHARANI\OneDrive\Desktop\Distributed System Lab Experiments>
```

RESULT:

Thus, the program to demonstrate message passing between two processes using Python was implemented successfully.

PRACTICAL. No: 4	Lamport Logical Clock Algorithm using Python

AIM:

To simulate Lamport Logical Clock Algorithm for ordering events in distributed systems using Python.

CODE:

```
p1 = int(input("Enter initial clock of P1: "))

p2 = int(input("Enter initial clock of P2: "))

p1 += 1

print("P1 Internal Event:", p1)

p1 += 1

msg = p1

print("P1 Sends Message at:", msg)

p2 = max(p2, msg) + 1

print("P2 Receives Message at:", p2)
```

OUTPUT:

```
PS C:\Users\R.BAVADHARANI\OneDrive\Desktop\Distributed System Lab Experiments> & C:/Users/R.BAVADHARANI/AppData/Local/Programs/Pyt
s/R.BAVADHARANI/OneDrive/Desktop/Distributed System Lab Experiments/Lamport-Clock/lamport.py"
Enter initial clock of P1: 0
Enter initial clock of P2: 0
P1 Internal Event: 1
P1 Sends Message at: 2
P2 Receives Message at: 3
```

RESULT:

Thus, the Lamport Logical Clock Algorithm was simulated successfully and event ordering was demonstrated.

PRACTICAL. No: 5	Distributed Sum of N Numbers using Python

AIM:

To implement distributed sum of N numbers by dividing data among nodes and calculating the total sum using Python.

CODE:

```
n = int(input("Enter number of elements: "))

nums = list(map(int, input("Enter elements: ").split()))

mid = n // 2

part1 = nums[:mid]

part2 = nums[mid:]

sum1 = sum(part1)

sum2 = sum(part2)

total = sum1 + sum2

print("First Node Data:", part1)

print("Second Node Data:", part2)

print("Partial Sum 1:", sum1)

print("Partial Sum 2:", sum2)

print("Total Sum:", total)
```

OUTPUT:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

s/R.BAVADHARANI/OneDrive/Desktop/Distributed System Lab Experiments/Distributed-sum/distributed sum.py"
Enter number of elements: 6
Enter elements: 1 2 3 4 5 6
First Node Data: [1, 2, 3]
Second Node Data: [4, 5, 6]
Partial Sum 1: 6
Partial Sum 2: 15
Total Sum: 21
```

RESULT:

Thus, the distributed sum of N numbers was implemented successfully and the total sum was obtained.

PRACTICAL. No: 6	Distributed Average / Mean Calculation

AIM: To calculate the average of numbers by dividing data into distributed parts and combining results.

CODE:

```
def part_average(data):
    return sum(data), len(data)
n = int(input("Enter number of elements: "))
data = list(map(int, input("Enter elements: ").split()))
parts = int(input("Enter number of parts: "))
size = n // parts
total_sum = 0
total_count = 0
for i in range(parts):
    part = data[i*size:(i+1)*size]
    s, c = part_average(part)
    total_sum += s
    total_count += c
avg = total_sum / total_count
print("Distributed Average:", avg)
```

OUTPUT:

```
TERMINAL

PS C:\Users\Asus\Desktop\Distributed system lab> & C:/Users/Asus/AppData/Local/Pr
buted system lab/Simple Compound Interest/Distributed Average.py"
Enter number of elements: 5
Enter elements: 10 15 20 25 30
Enter number of parts: 4
Distributed Average: 17.5
PS C:\Users\Asus\Desktop\Distributed system lab>
```

RESULT: The average was successfully calculated using distributed partitions.

PRACTICAL. No: 7	Multi-threaded Server Handling Multiple Clients

AIM: To develop a multi-threaded server that can handle multiple client requests simultaneously using sockets in Python.

CODE:

SERVER.PY

```
import socket
import threading
def handle_client(conn, addr):
    print("Connected:", addr)
    msg = conn.recv(1024).decode()
    print("Client:", msg)
    conn.send("Message received".encode())
    conn.close()
server = socket.socket()
server.bind(("localhost", 5000))
server.listen(5)
print("Server started...")
while True:
    conn, addr = server.accept()
    thread = threading.Thread(target=handle_client, args=(conn, addr))
    thread.start()
```

CLIENT.PY

```
import socket

client = socket.socket()

client.connect(("localhost", 5000))

msg = input("Enter message: ")

client.send(msg.encode())

print(client.recv(1024).decode())

client.close()
```

OUTPUT:

```
TERMINAL

PS C:\Users\Asus\Desktop\Distributed system lab> & C:/Users/Asus/AppData/Local/Prog
buted system lab/multiple server handles multiple client/Server.py"
Server started...
Connected: ('127.0.0.1', 52247)
Client: hii
Connected: ('127.0.0.1', 52253)
Client: ds
Connected: ('127.0.0.1', 64847)
Client: DFHDHHTD
```

```
TERMINAL

PS C:\Users\Asus\Desktop\Distributed system lab> & C:/Users/Asus/AppData/Loca
buted system lab/multiple server handles multiple client/Client.py"
Enter message: hii
Message received
PS C:\Users\Asus\Desktop\Distributed system lab> & C:/Users/Asus/AppData/Loca
buted system lab/multiple server handles multiple client/Client.py"
Enter message: ds
Message received
PS C:\Users\Asus\Desktop\Distributed system lab>
```

RESULT:

The multi-threaded server was successfully implemented, and it handled multiple clients concurrently using threads.

PRACTICAL. No: 8	Distributed Chat Application using Sockets

AIM: To implement a two-way communication chat system using sockets.

CODE:

SERVER.PY

```
import socket
server = socket.socket()
server.bind(("localhost", 6000))
server.listen(1)
conn, addr = server.accept()
print("Connected:", addr)
while True:
    msg = conn.recv(1024).decode()
    print("Client:", msg)
    reply = input("You: ")
    conn.send(reply.encode())
```

CLIENT.PY

```
import socket
client = socket.socket()
client.connect(("localhost", 6000))
while True:
    msg = input("You: ")
    client.send(msg.encode())
    reply = client.recv(1024).decode()
    print("Server:", reply)
```

OUTPUT:

```
TERMINAL

PS C:\Users\Asus\Desktop\Distributed system lab> & C:/Users/Asus/AppData/Local/Programs/Python/Pyt
buted system lab/DISTRIBUTED CHAT APPLICATION/Server.py"
Connected: ('127.0.0.1', 64112)
Client: hi
You: hello
```

```
TERMINAL

PS C:\Users\Asus\Desktop\Distributed system lab> & C:/Users/Asus/AppData/Local/P
buted system lab/DISTRIBUTED CHAT APPLICATION/client.py"
You: hi
Server: hello
You:
```

RESULT:

Two-way communication between client and server was achieved successfully.

PRACTICAL. No: 9	**Bully Leader Election Algorithm**

AIM: To elect a leader using Bully Algorithm where the highest process ID becomes leader.

CODE:

```
processes = list(map(int, input("Enter process IDs: ").split()))
failed = int(input("Enter failed process ID: "))
active = [p for p in processes if p != failed]
initiator = int(input("Enter initiator process: "))
higher = [p for p in active if p > initiator]
if not higher:
    leader = initiator
else:
    leader = max(active)
print("Elected Leader:", leader)
```

OUTPUT:

```
TERMINAL
PS C:\Users\Asus\Desktop\Distributed system lab> & C:/Users/Asus/AppData/Local/
ystem lab/Bully election.py"
Enter process IDs: 1 2 3 4 5
Enter failed process ID: 4
Enter initiator process: 3
Elected Leader: 5
PS C:\Users\Asus\Desktop\Distributed system lab>
```

RESULT:

Leader election was successfully performed using Bully Algorithm.

PRACTICAL. No: 10	Ring-based Leader Election Algorithm

AIM: To elect a leader in a ring topology where processes pass messages in order.

CODE:

```
processes = list(map(int, input("Enter process IDs: ").split()))
n = len(processes)
initiator = int(input("Enter initiator index (0-based): "))
msg = []
i = initiator
while True:
    msg.append(processes[i])
    i = (i + 1) % n
    if i == initiator:
        break
leader = max(msg)
print("Processes in ring:", msg)
print("Elected Leader:", leader)
```

OUTPUT:

```
TERMINAL
PS C:\Users\Asus\Desktop\Distributed system lab> & C:/Users/Asus/AppData/Local/Prog
ystem lab/ring election.py"
Enter process IDs: 1 2 3 4 5
Enter initiator index (0-based): 3
Processes in ring: [4, 5, 1, 2, 3]
Elected Leader: 5
PS C:\Users\Asus\Desktop\Distributed system lab>
```

RESULT:

Leader was successfully elected using ring-based algorithm.